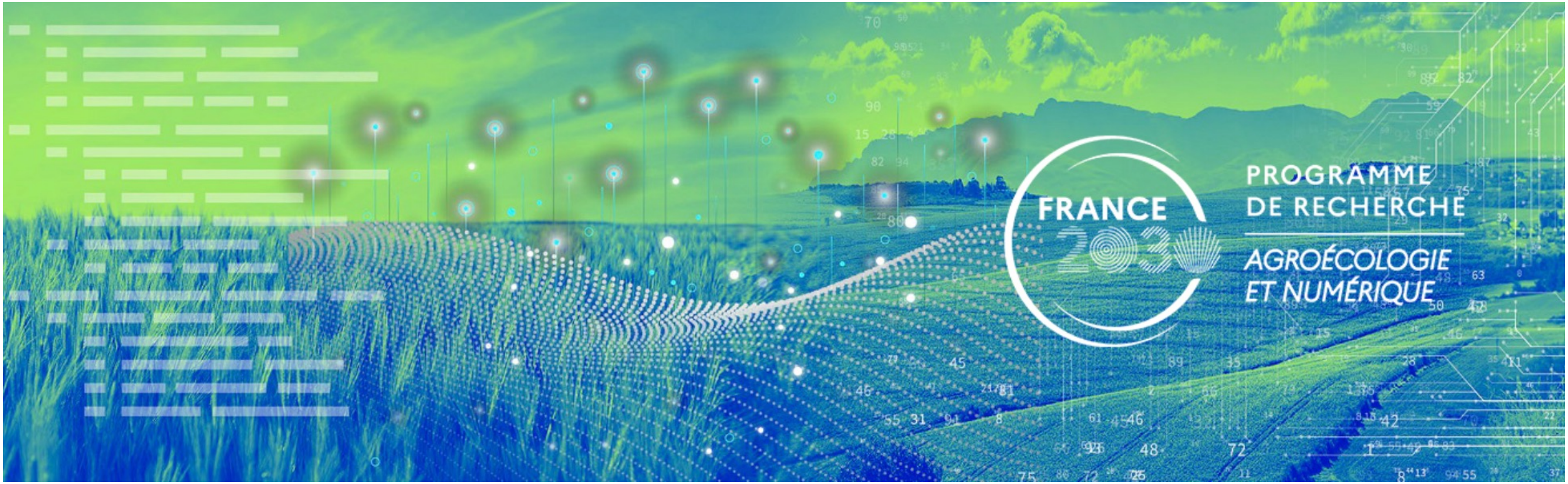

Building the LoRa IoT platform
Part 2: update for PCB v5 (RAK3172)

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List of components

Image shown is the PCBA v5 from INTEL-IRRIS

For SOLAR

For 2xAA alkaline batteries

For SOLAR with NiMh battery

SMA male

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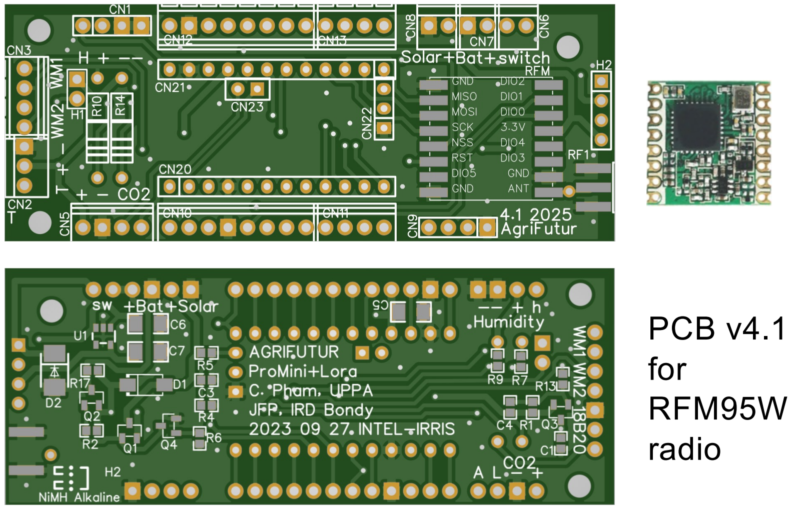
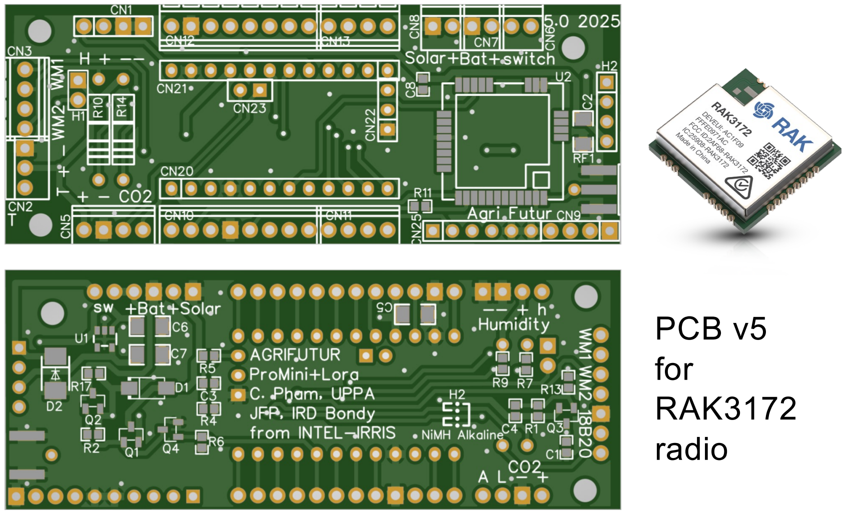
Important

- This tutorial is an update for the PCB v5 based on the RAK3172 radio module
- Reader MUST first look at the tutorial presenting the PCB v4.1 to understand the common instructions
- This tutorial only presents the differences between the 2 versions



PCB v4.1 for RFM95W radio



PCB v5 for RAK3172 radio

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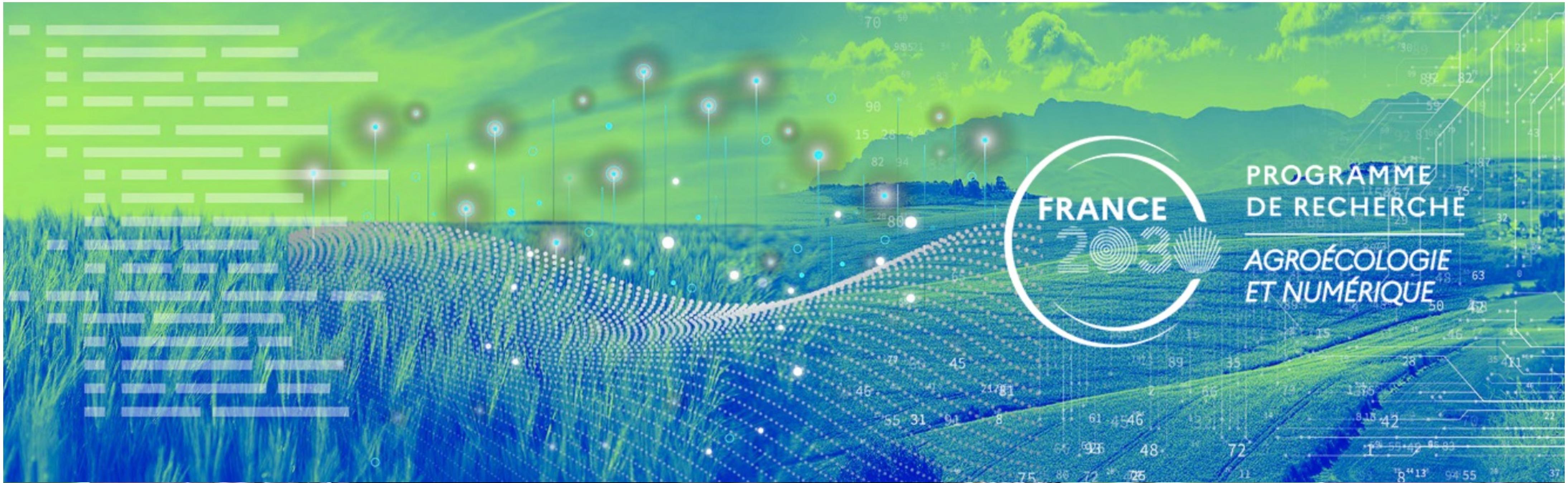
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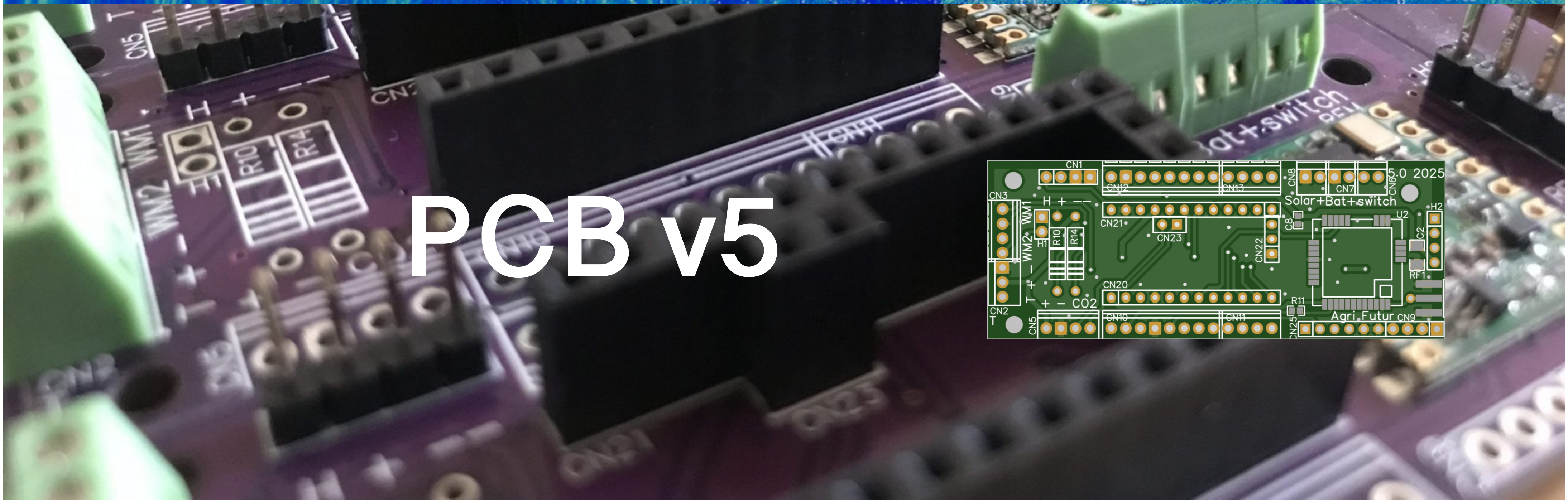
PCB v5
5.0 2025
Solar+Bat+switch
Agri.Futur

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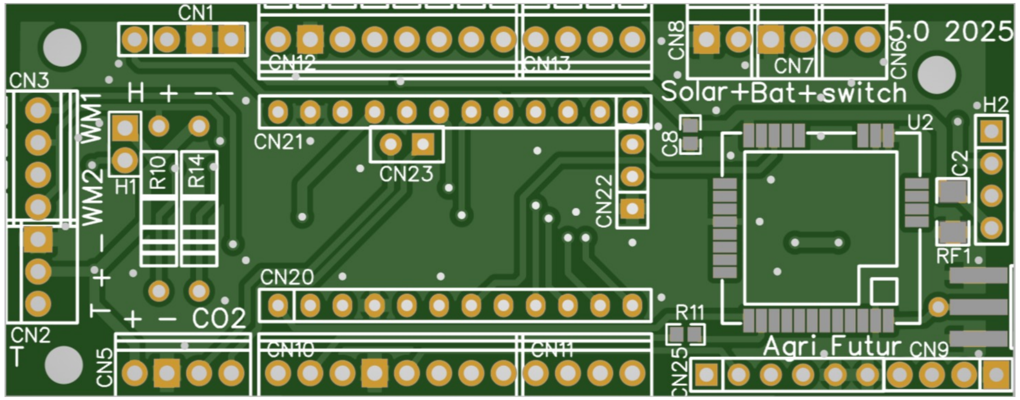
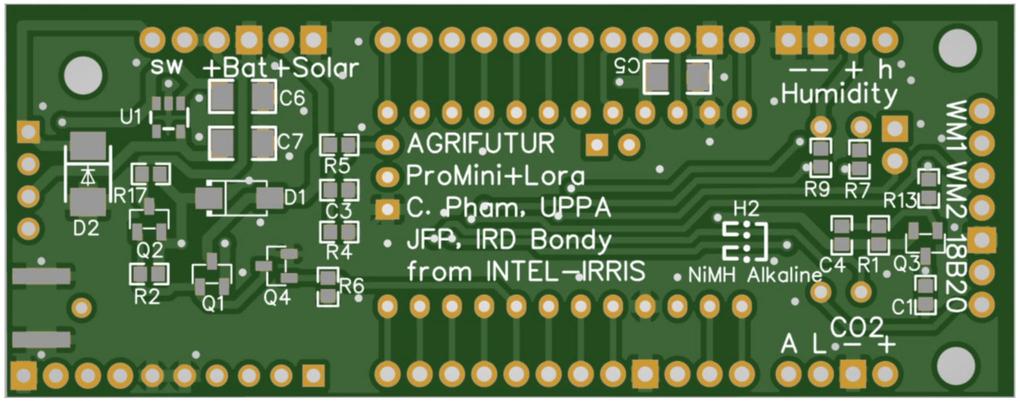
The PCB v5 - raw version

- You can order the raw version of the PCB, which means only the PCB, without any electronic components soldered by the manufacturer
- It is the so-called DIY approach
- The raw version is not intended to have solar charging capabilities





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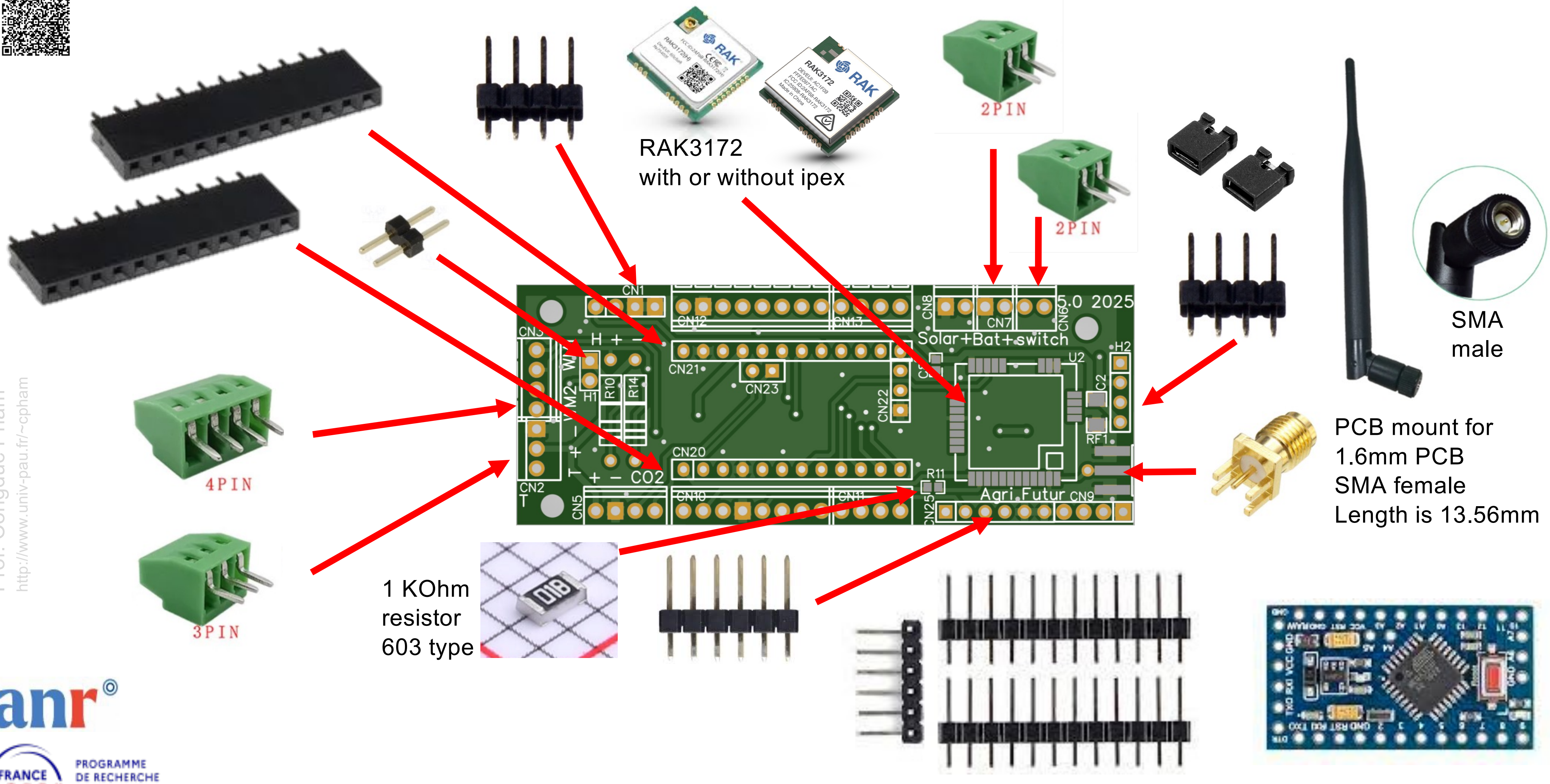
Electronic parts - starter-kit version



Arduino ProMini 3.3V 8MHz

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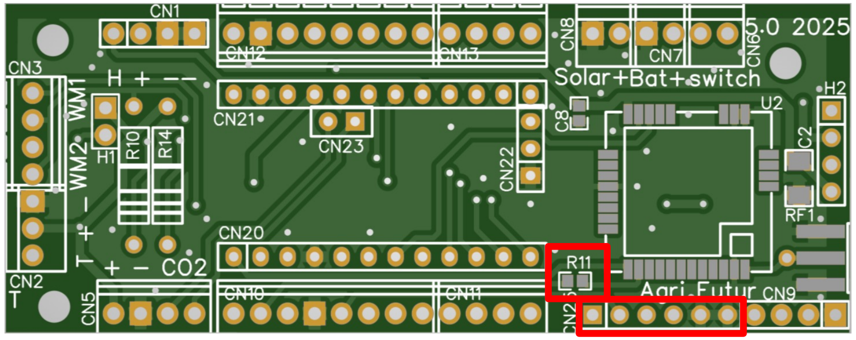
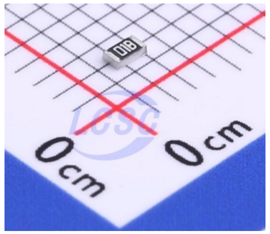
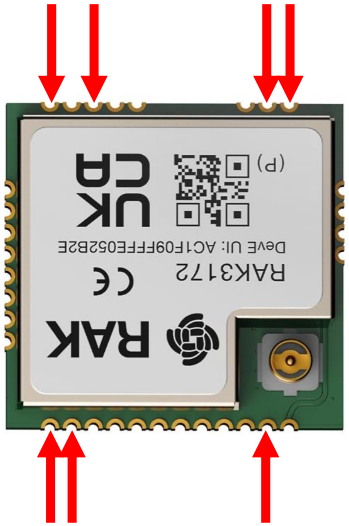
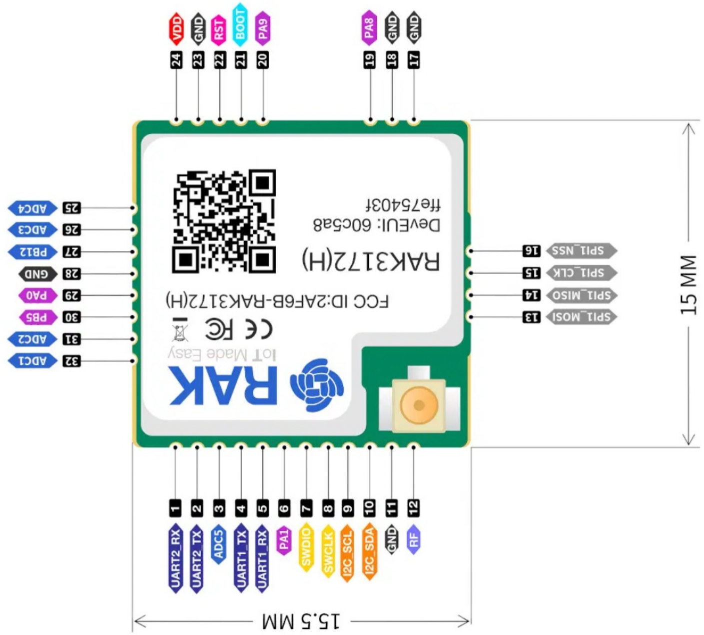
Difference compared to PCB v4.1

- The RAK3172 radio replaces the RFM95W radio module. No need to solder all pins, only those marked by the red arrows

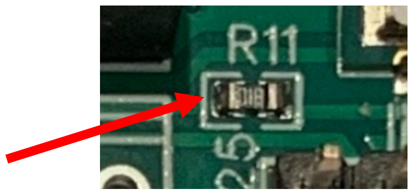








- A 6-pin header needs to be soldered in order to enable serial monitor debug from the Arduino IDE
- A 1KOhm resistor must be soldered at R11 place



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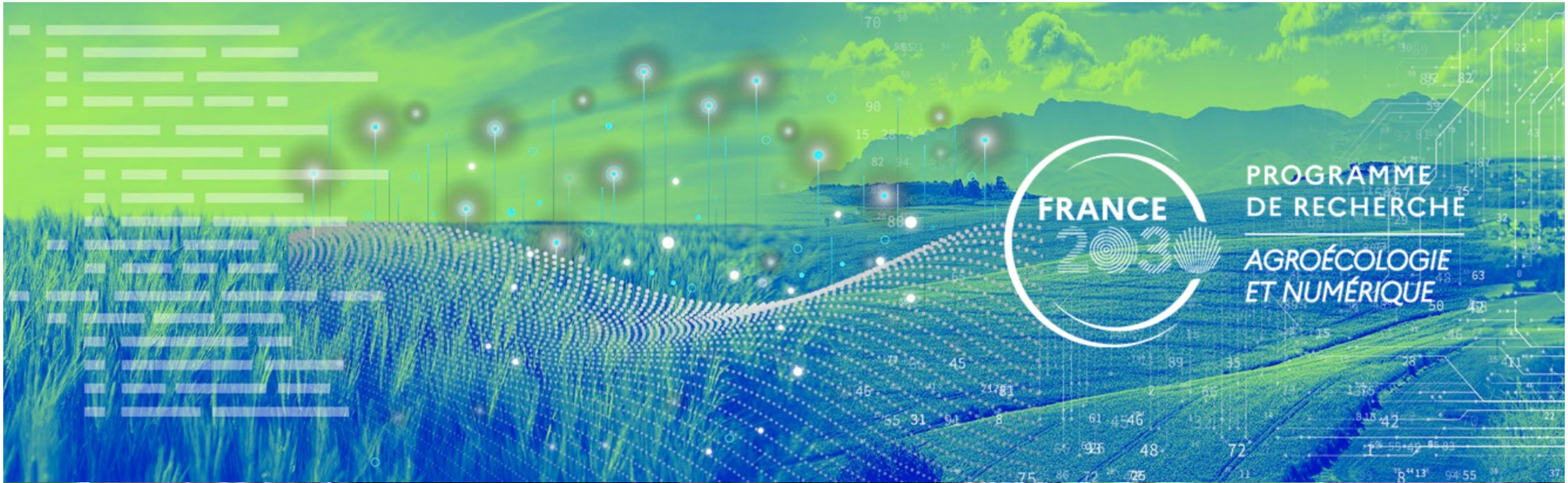
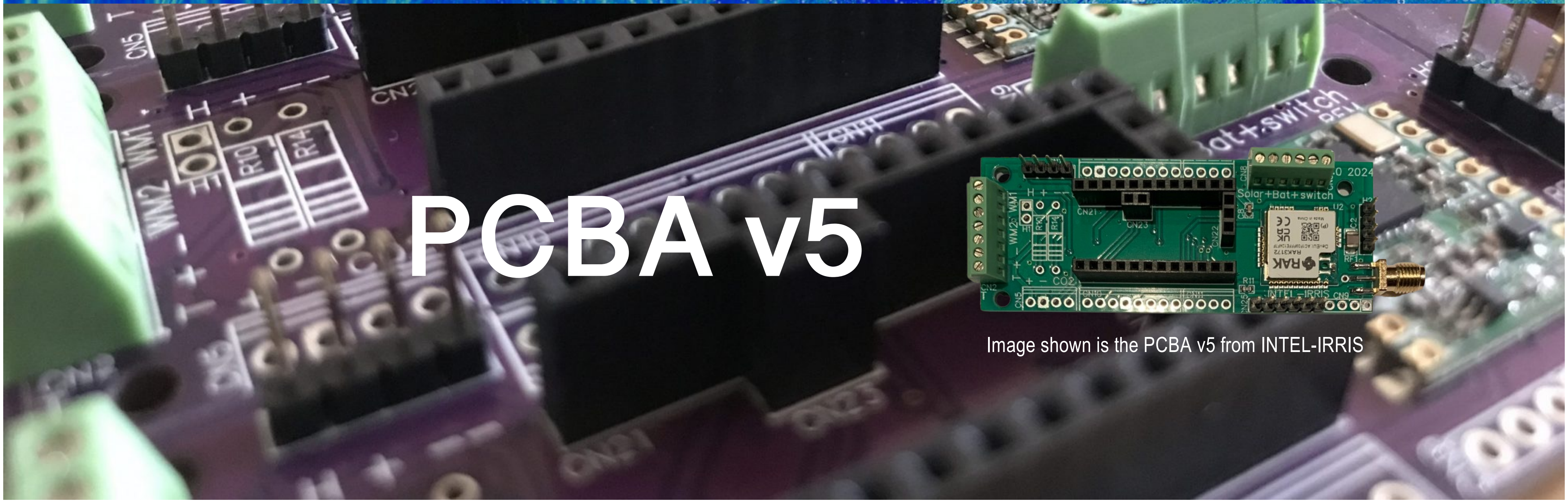


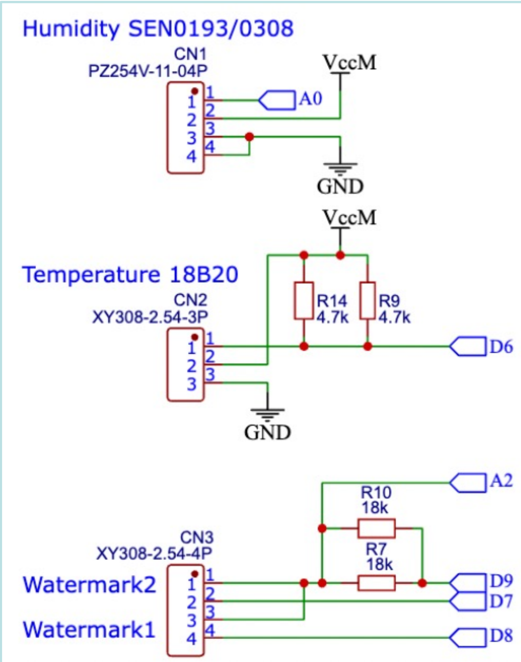


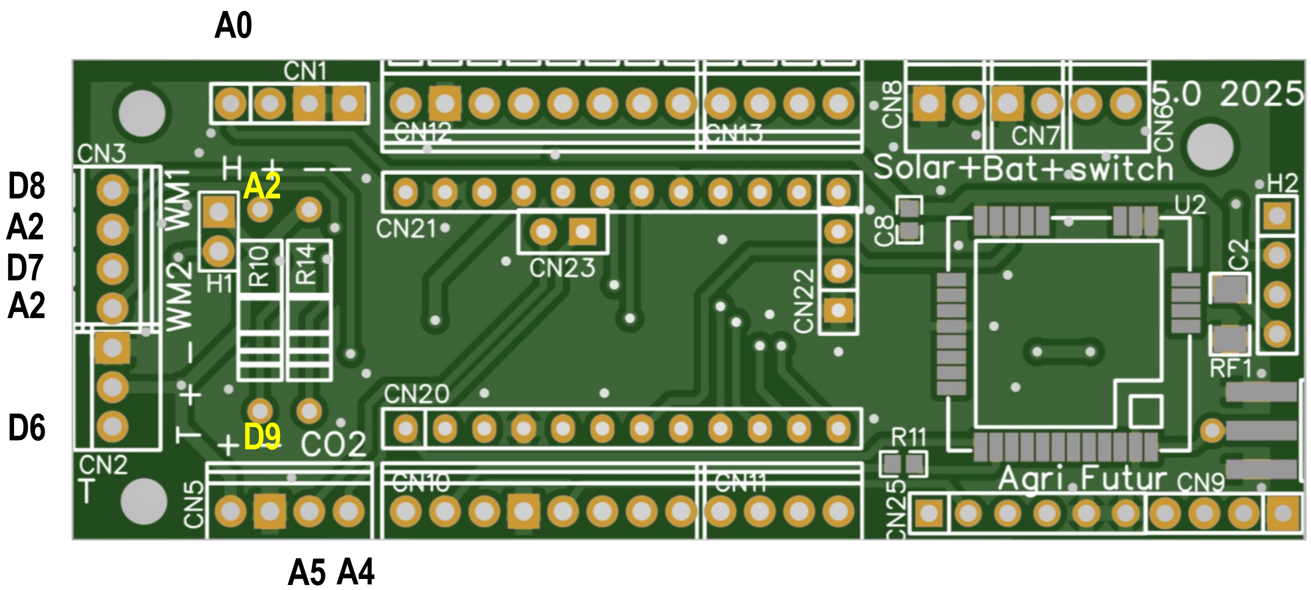
PCBA v5

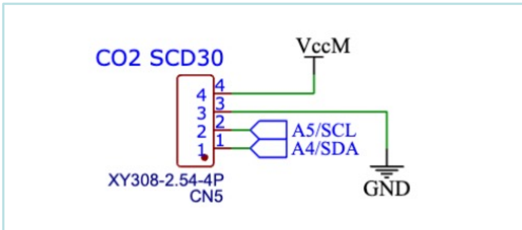
Image shown is the PCBA v5 from INTEL-IRRIS

PCB/PCBA v5 pin description

- The PCB has the following pre-defined wiring to Arduino ProMini pin
  https://github.com/CongducPham/PEPR_AgriFutur/blob/main/PCBs/IRD_PCB_5/Schematic_IISS_5_2024-05-06.pdf







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Soil device with PCBA v5, final result

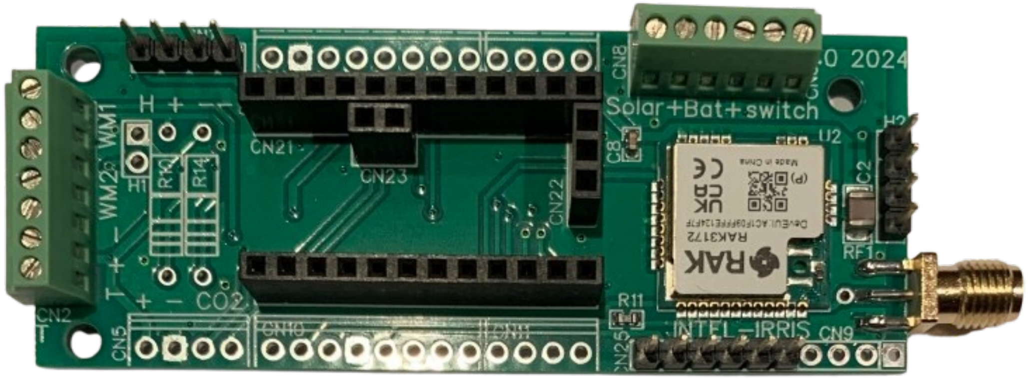
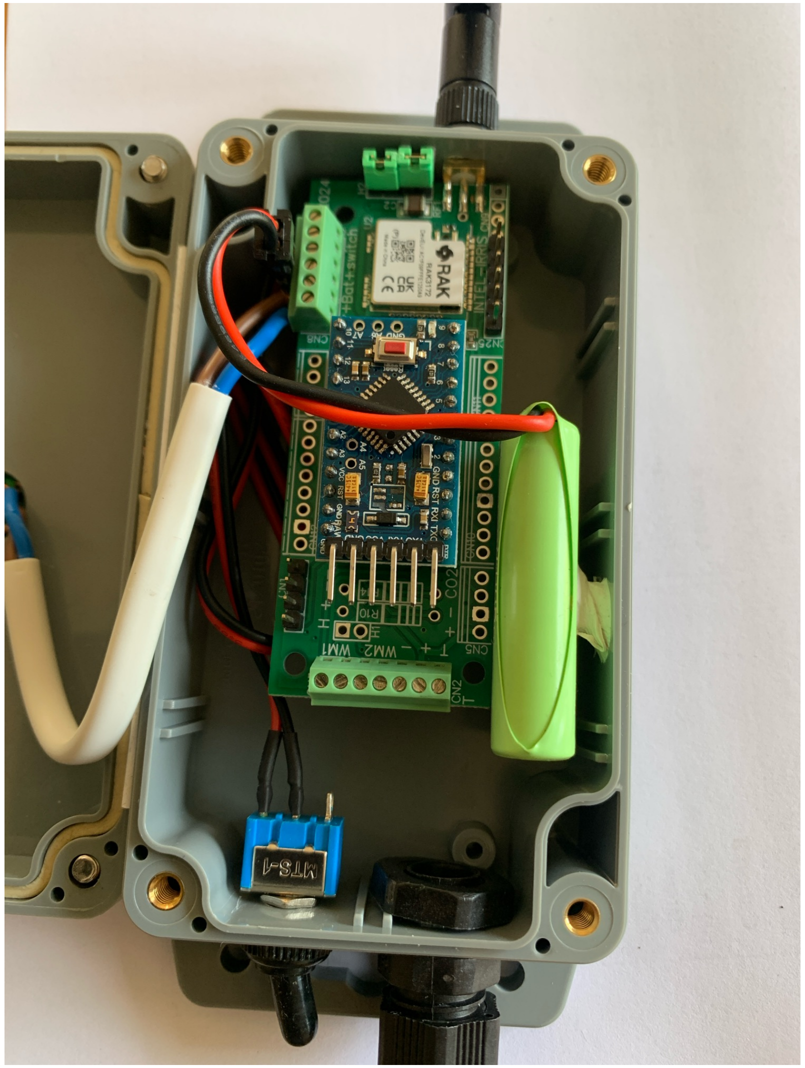


Image shown is the PCBA v5 from INTEL-IRRIS



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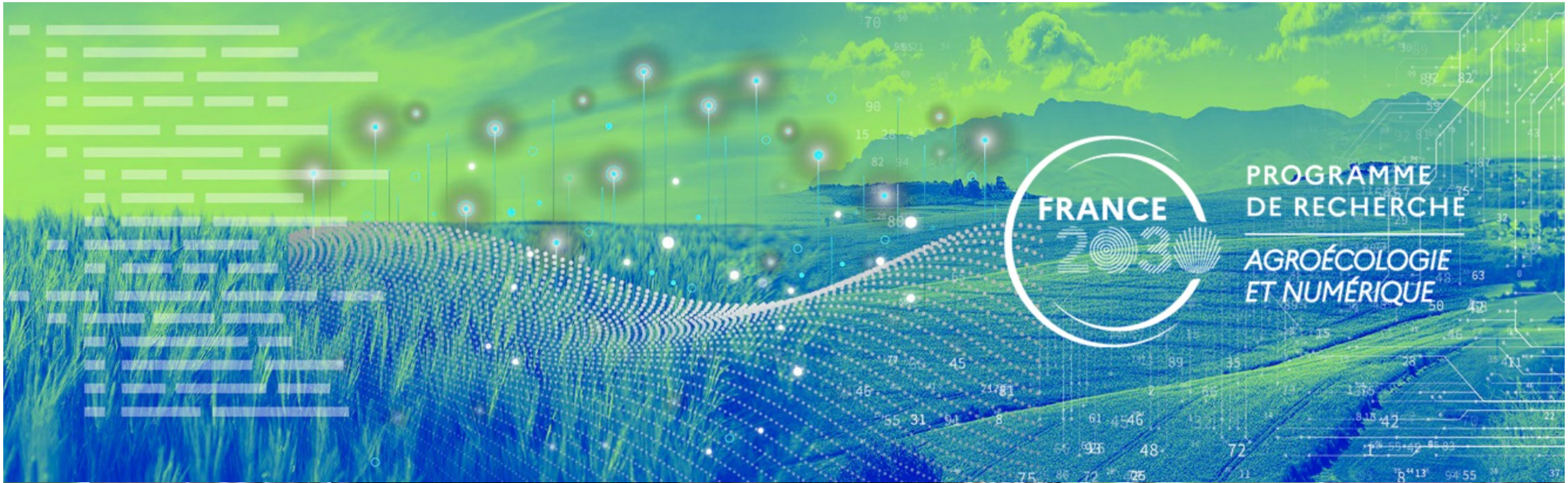

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Configuration for PCB/PCBA v5 (1)

- For raw version, uncomment `IRD_PCB` in BoardSettings.h
- For PCBA version, also uncomment `IRD_PCBA`

```
////////////////////////////////////////////////////////////////////
// uncomment for IRD PCB board (both v4.1 and v5)
#define IRD_PCB
// also uncomment for IRD PCB board fully assembled by manufacturer
// with all components, including solar circuit (both v4.1 and v5)
#define IRD_PCBA
```

- ONLY for solar version, uncomment `SOLAR_BAT`

```
////////////////////////////////////////////////////////////////////
// uncomment only if the IRD PCB or PCBA is running on solar panel
// MUST be commented if running on alkaline battery
// code for SOLAR_BAT has been written by Jean-François Printanier from IRD
#define SOLAR_BAT
// do not change if you are not knowing what you are doing
#define NIMH
```

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Configuration for PCB/PCBA v5 (2)

- Uncomment `SOFT_SERIAL_DEBUG` in BoardSettings.h

```
///////////////////////////////////////////
// uncomment for IRD PCB RAK version where
#define SOFT_SERIAL_DEBUG
```

- Uncomment `RAK3172` in RadioSettings.h

```
/////////////////////////////////////
// please uncomment only 1 choice
//#define SX126X
//#define SX127X
//#define SX128X
#define RAK3172
/////////////////////////////////////
```

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Serial monitor with PCB v5

- After flashing the Arduino, it is usually desirable to watch at the text output from the Arduino IDE serial monitor to check that everything went OK
- With PCB v5, the RAK radio module is controlled by AT commands received on its serial port and sent by the Arduino on its serial port (baud rate is normally set to 38400)
- Therefore, Arduino's TX/RX pins are connected to the RAK3172 and the serial monitor used traditionally to look at text output can not use the default Arduino TX/RX pins
- The PCB v5 uses a software defined serial port for the text output where only TX is defined on digital pin 2

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Connect the FTDI USB-Serial

- After flashing the Arduino with the FTDI USB-Serial cable (left photo), unplug it in order to plug it again on the dedicated debug header on the PCB v5 (middle & right photo)

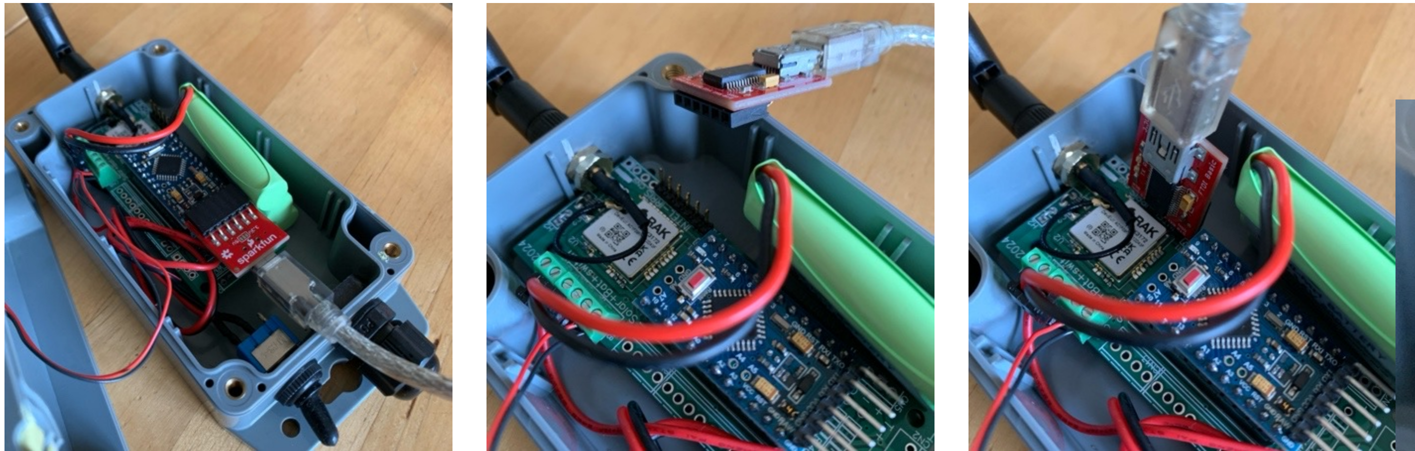
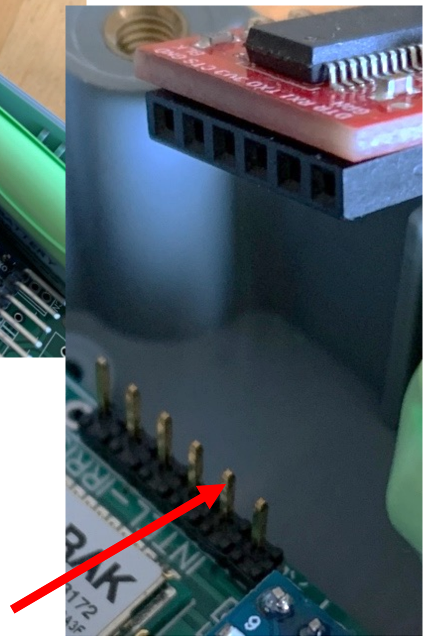


Image shown is the PCBA v5 from INTEL-IRRIS



- Battery should be connected, but switch is on OFF
- Be sure to connect the RX pin of the FTDI USB-Serial to the TX pin of the debug header on the PCB v5

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Getting text output from the board

- Order of steps is important!

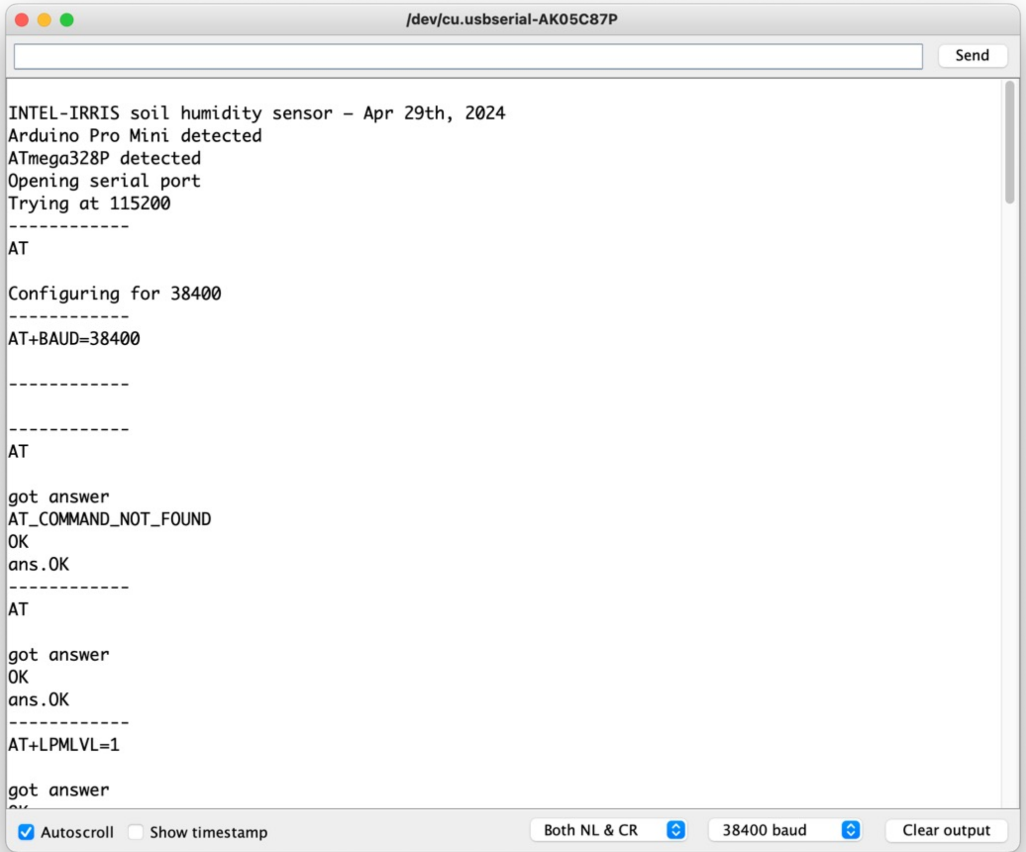

Image shown is the Arduino IDE with INTEL-IRRIS code

Open serial monitor, you should see nothing

Set baud rate to 38400

Switch ON the board

See output from board

Check that transmission is OK

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Configuration summary: check list

- PCBA v5 →
  - comment SX127X
  - uncomment SOFT_SERIAL_DEBUG and RAK3172
- RAW PCB v5 → comment IRD_PCBA
- No solar panel → default configuration, nothing to do ✔
- Solar panel → uncomment SOLAR_BAT, typically use 3xAAA NiMh
- See "Part 1: sensor device with PCB v4.1" for sensor configuration

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Annex: RAK3172 baud rate

- Latest version of RAK3172 is using by default (factory setting) a baud rate of 115200, which is too fast (mostly for RX) for the Arduino Pro mini based on ATMega328P microcontroller
- The device code is actually reprogramming the RAK3172 to use a baud rate of 38400, therefore you do not need to worry about setting the correct baud rate on the RAK3172, especially with the fully assembled version of the PCB where the RAK3172 is provided and assembled on the PCB by the PCB manufacturer

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Annex: IPEX or not IPEX?

- The BOM file for the PCBA fully assembled is using the RAK3172 without the IPEX connector because we prefer to have a robust soldered RF connector

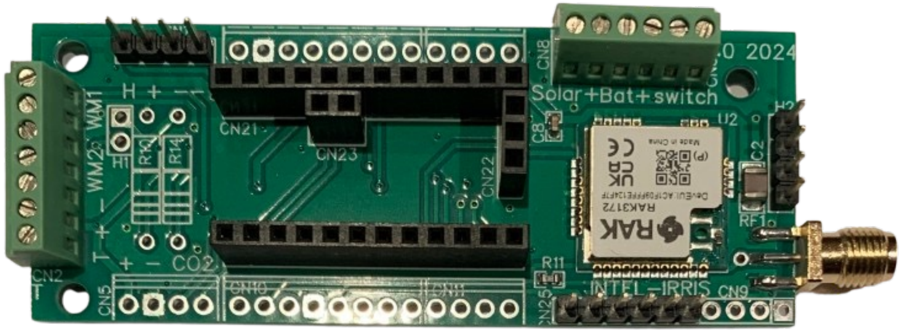
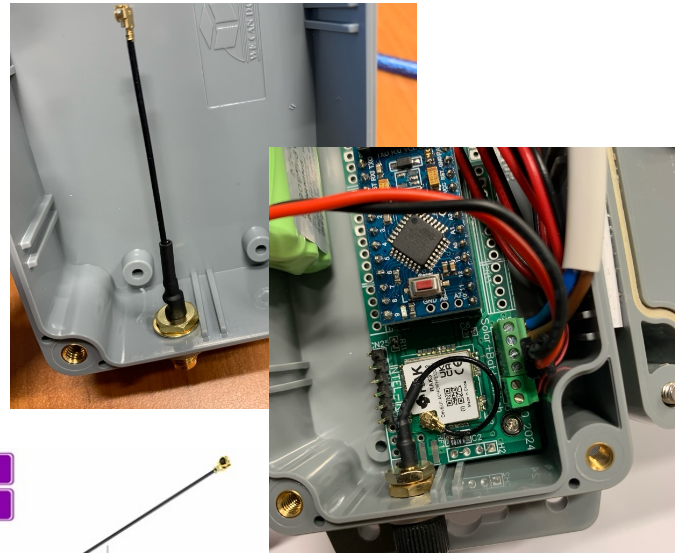
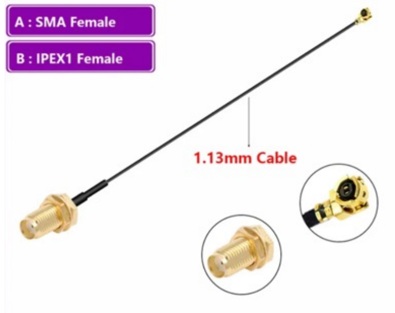


Image shown is the PCBA v5 from INTEL-IRRIS



- For the raw version of the PCB, you can solder the version with IPEX and connect the antenna with an IPEX-female to SMA female short pigtail



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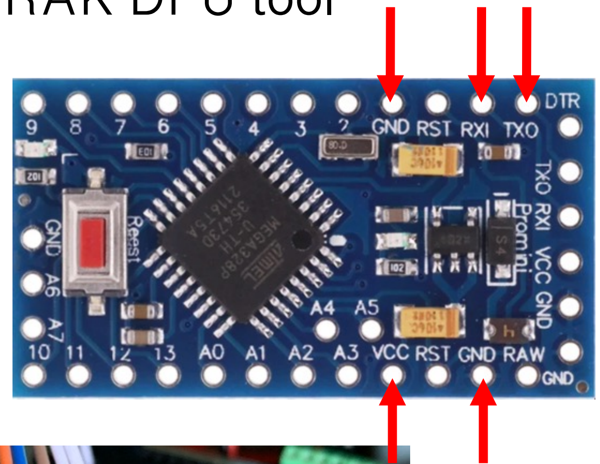
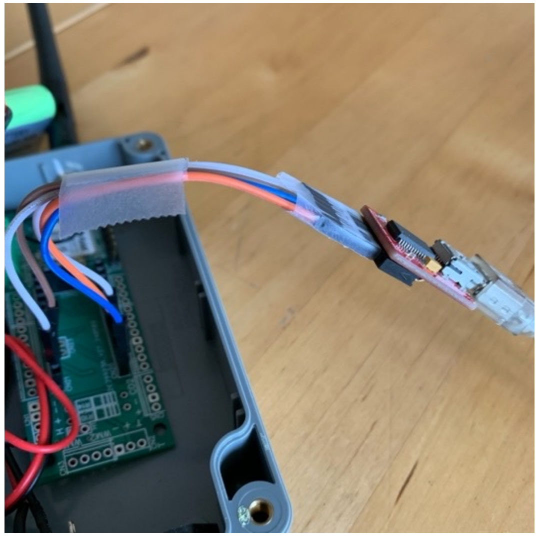
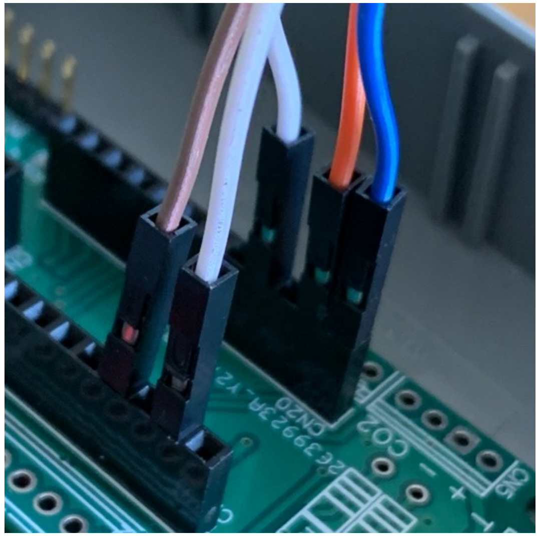
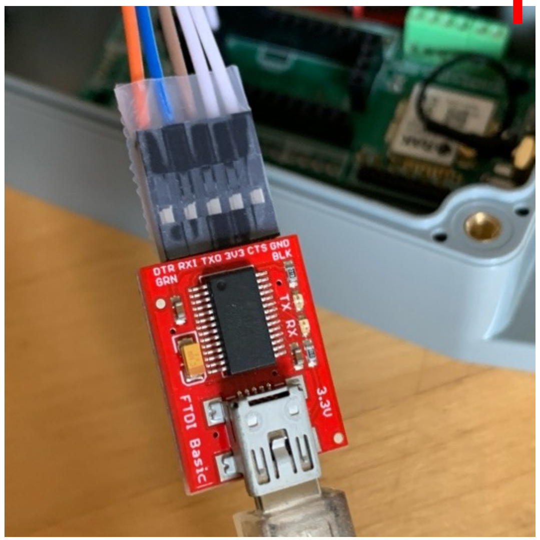
Annex: update firmware of RAK3172

- The RAK3172's firmware can be upgraded with the RAK DFU tool
- You can use Dupont wires to directly connect VCC, GND, RX and TX of the FTDI USB-serial cable to corresponding pins on the female headers after removing the Arduino Pro Mini from its socket. DTR is not connected









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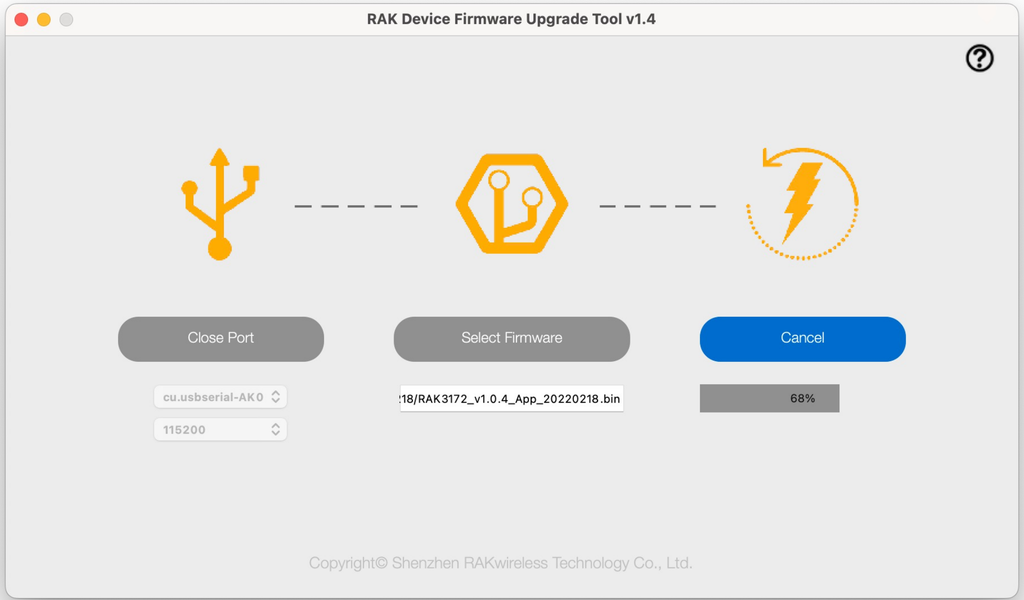
Annex: RAK DFU tool

- Get latest RAK3172 firmware from https://downloads.rakwireless.com/#RUI/RUI3/Image/ usually, it is the RAK3172-E_latest.bin file
- Use RAK DFU tool to flash the new firmware



You may need to issue AT+BOOT from a serial tool in order to be able to upgrade with DFU tool

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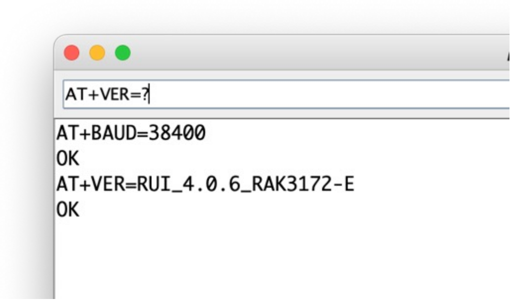
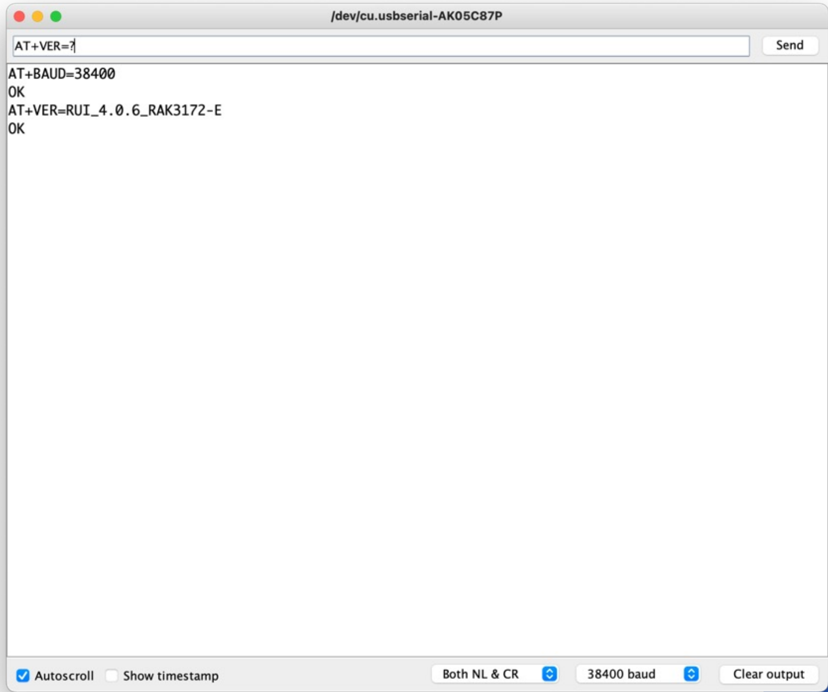
Annex: send direct AT commands

- With the FTDI USB-serial cable connected to the PCB v5 without the Arduino Pro Mini, as show on the previous slide, you can directly communicate with the RAK3172 to send AT commands for test purposes
- Any serial communication tool can be used, including the Arduino IDE serial monitor
- RAK also proposes the WisToolBox software to communicate with RAK components





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Annex: AT commands at startup

- Here is the list of AT commands used by the device
- During setup(), for ABP mode in EU868 band, for capacitive
  - AT+BAUD=38400
  - AT+LPMLVL=1
  - AT+LPM=0
  - AT+LPM=? ; AT+VER=? ; AT+BAND=? ; AT+ADR=? ; AT+NJM=? ; AT+DEVADDR=?; AT+NWKSKEY=? ; AT+APPSKEY=?
  - AT+NJM=0 ; AT+CLASS=A ; AT+BAND=4 ; AT+ADR=0
  - AT+DEVADDR=26011DAA
  - AT+NWKSKEY=23158D3BBC31E6AF670D195B5AED5525
  - AT+APPSKEY=23158D3BBC31E6AF670D195B5AED5525
  - AT+LPMLVL=2
  - AT+LPM=1
- On wake up, to transmit (e.g. sending 898 and 3.45, LPP format)
  - AT+SEND=1:0067231406020159
  - AT+SLEEP

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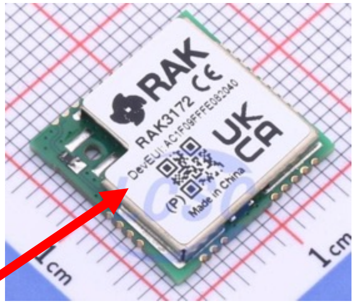
Annex: OTAA with RAK3172 (1)

- The device with the RAK3172 can be integrated into a LoRaWAN ecosystem and configured by OTAA (Over The Air Activation) to dynamically get its address & session keys
- Uncomment `ENABLE_OTAA` in native_at_lorawan.h

```
#include "BoardSettings.h"

#define ENABLE_OTAA
```



- When using OTAA, you need to know
  - DevEUI: unique identifier of the device, it is usually pre-provisioned by the radio module manufacturer and indicated on the radio module
  - AppEUI: or JoinEUI, it is usually pre-provisioned by the radio module manufacturer. For RAK, it is usually AC1F09FFF8683172
  - AppKey: it is usually pre-provisioned by the radio module manufacturer. For RAK it is usually DevEUI+AppEUI

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Annex: OTAA with RAK3172 (2)

- It is possible to use the default values pre-provisioned by the manufacturer for DevEUI, AppEUI and AppKey IF YOU DIDN'T UPGRADE THE FIRMWARE
  - Leave `OVERWRITE_OTAA_EUI` commented
- If RAK3172 firmware has been updated, they may not be set anymore! In this case, uncomment `OVERWRITE_OTAA_EUI` in native_at_lorawan.cpp and set in DevEUI, AppEUI and AppKey
- Flash device and run once to set DevEUI, AppEUI and AppKey in the RAK module
- Then, comment `OVERWRITE_OTAA_EUI` to flash again and run

```
////////////////////////////////////////////////////////////////////
//ENTER HERE your device info (same order, i.e. msb)
////////////////////////////////////////////////////////////////////
//#define OVERWRITE_OTAA_EUI

#ifdef OVERWRITE_OTAA_EUI
//DevEui is normally indicated on the radio module
char* DevEuiStr="AC1F09FFFE12AAAA";
//here is the default used by RAK
char* AppEuiStr="AC1F09FFF8683172";
//here is the default used by RAK: DEVEUI+APPEUI
char* AppKeyStr="AC1F09FFFE12AAAAAC1F09FFF8683172";
#endif
```

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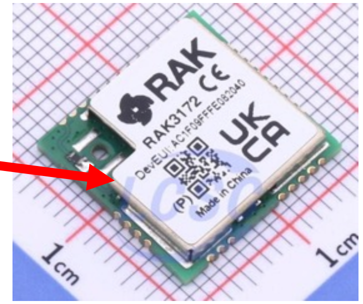
Annex: register an OTAA device

- To register your RAK3172 OTAA device on a LoraWAN Network Server such as TheThingNetwork (TTN) you can then just enter
  - DevEUI: get it from the sticker
    For instance AC1F09FFFE12AAAA
  - AppEUI or JoinEUI: it is AC1F09FFF8683172
  - AppKey: should be AC1F09FFFE12AAAAAC1F09FFF8683172



- That's all. Once powered your device will join and you should receive data if a LoRaWAN gateway is near your device
- IMPORTANT, the gateway is not full LoRaWAN in its single-channel version
- We will discuss how to get data from an OTAA device, locally on the gateway in a dedicated tutorial

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